

Finite-dimensional bistochastic quantum random variables are classical

Candidate partial result for arXiv:2005.00005

2026-06-21

Status

This packet records a candidate partial result, likely valid. It resolves the finite-dimensional Hilbert-space case of a question raised by Plosker and Ramsey in [1]. It is not claimed to settle the full separable Hilbert-space formulation.

Source question

The source paper is Sergei Plosker and Brian Ramsey, “Bistochastic operators and quantum random variables”, arXiv:2005.00005 [1]. In Section 5, after Definition 5.1 and Theorem 5.5, the authors show that every classical bistochastic operator on $L^1(X, \mu)$ extends entrywise to a bistochastic operator on $L^1_{\mathcal{H}}(X, \mu I_{\mathcal{H}})$. Immediately afterwards they write:

again in Section 6, which in fact comes automatically with the definition of B , is $(Bf)_s = B(f_s)$. □

We will refer to the extension developed in the previous theorem by B as well and the set of such bistochastic operators as $\mathfrak{B}(L^1(X, \mu))$ still. We have no example of a bistochastic operator on $L^1_{\mathcal{H}}(X, \mu I_{\mathcal{H}})$ that does not arise in this way.

Corollary 5.6. *If $\nu = \mu I_{\mathcal{H}}$ for some finite, positive measure μ , then for every $B \in \mathfrak{B}(L^1(X, \mu))$, $f \in L^1(X, \mu)$, $A \in \mathcal{B}(\mathcal{H})$ and $g \in L^\infty(X, \mu)$*

In source-line form, the key sentence is line 1088 of the parsed source:

“We have no example of a bistochastic operator on $L^1_{\mathcal{H}}(X, \mu I_{\mathcal{H}})$ that does not arise in this way.”

Definitions

Let (X, Σ, μ) be a finite measure space and let $\nu = \mu I_{\mathcal{H}}$. We use Plosker–Ramsey’s definition: a linear operator B on $L^1_{\mathcal{H}}(X, \nu)$ is *bistochastic* if

$$B \geq 0, \quad \int_X Bf \, d\mu = \int_X f \, d\mu \quad (f \in L^1_{\mathcal{H}}(X, \nu)), \quad BI_{\mathcal{H}} = I_{\mathcal{H}}.$$

The source paper proves that a classical bistochastic operator S on $L^1(X, \mu)$ gives such a B by

$$B(fA) = S(f)A, \quad f \in L^1(X, \mu), \quad A \in \mathcal{B}(\mathcal{H}).$$

Idea of the proof

The obstruction to exotic examples is already present on indicator functions. Given two measurable sets E, F , integrate $B(\chi_E A)$ over F . This produces a positive linear map $\Theta_{F,E} : M_d \rightarrow M_d$. Since B preserves the full integral, $\Theta_{X,E} = \mu(E) \text{id}$, so $\Theta_{F,E}$ lies in the positive order interval $[0, \mu(E) \text{id}]$.

In finite dimensions that order interval is one-dimensional: every positive map $M_d \rightarrow M_d$ dominated by a scalar multiple of the identity map is itself a scalar multiple of the identity. Rank-one projections and the two-by-two matrix-unit corners give an elementary proof. Therefore every $\Theta_{F,E}$ is scalar, and the scalar coefficients form the transition measures of an ordinary Markov/bistochastic operator S on $L^1(X, \mu)$. The identity $B(\chi_E A) = S(\chi_E)A$ then extends from indicators to simple functions and, by density in finite dimension, to all of $L^1_{\mathbb{C}^d}(X, \mu I_d)$.

Lemma 1 (Order interval of the identity on matrices). *Let $d < \infty$, and let $\Phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$ be a positive linear map. If $c \text{id} - \Phi$ is positive for some $c \geq 0$, then $\Phi = \lambda \text{id}$ for some $0 \leq \lambda \leq c$.*

Proof. If $c = 0$ the claim is immediate, so scale to $c = 1$. Let $(e_i)_{i=1}^d$ be the standard basis and let E_{ij} be the matrix units. For each i , the rank-one projection E_{ii} satisfies

$$0 \leq \Phi(E_{ii}) \leq E_{ii}.$$

Hence $\Phi(E_{ii}) = \lambda_i E_{ii}$ for some $0 \leq \lambda_i \leq 1$.

Fix $i \neq j$. Put

$$p_{\pm} = \frac{1}{2}(E_{ii} + E_{jj} \pm E_{ij} \pm E_{ji}),$$

the projections onto the lines spanned by $e_i \pm e_j$. Since $0 \leq \Phi(p_{\pm}) \leq p_{\pm}$, there are scalars $\alpha_{\pm} \in [0, 1]$ such that $\Phi(p_{\pm}) = \alpha_{\pm} p_{\pm}$. Adding the two identities and using $p_+ + p_- = E_{ii} + E_{jj}$ gives

$$\lambda_i E_{ii} + \lambda_j E_{jj} = \frac{\alpha_+ + \alpha_-}{2}(E_{ii} + E_{jj}) + \frac{\alpha_+ - \alpha_-}{2}(E_{ij} + E_{ji}).$$

Therefore $\alpha_+ = \alpha_-$ and $\lambda_i = \lambda_j$. Since i, j were arbitrary, all λ_i 's are a common scalar λ .

The same identity with p_+ now yields

$$\Phi(E_{ij} + E_{ji}) = \lambda(E_{ij} + E_{ji}).$$

Using the rank-one projections onto $e_i \pm ie_j$ gives

$$\Phi(iE_{ij} - iE_{ji}) = \lambda(iE_{ij} - iE_{ji}).$$

Combining the last two displayed identities gives $\Phi(E_{ij}) = \lambda E_{ij}$ and $\Phi(E_{ji}) = \lambda E_{ji}$. Thus Φ is λid on all matrix units. $\square$

Theorem 1. *Let (X, Σ, μ) be a finite measure space and let $\mathcal{H} = \mathbb{C}^d$. If B is a bistochastic operator on $L^1_{\mathcal{H}}(X, \mu I_d)$ in the sense of [1], then there is a unique classical bistochastic operator S on $L^1(X, \mu)$ such that*

$$B(fA) = S(f)A \quad (f \in L^1(X, \mu), A \in M_d(\mathbb{C})).$$

Consequently, for finite-dimensional $\mathcal{H}$, every bistochastic operator on $L^1_{\mathcal{H}}(X, \mu I_{\mathcal{H}})$ is one of the entrywise classical extensions constructed in [1, Theorem 5.5].

Proof. Let $E, F \in \Sigma$. For $A \in M_d$, define

$$\Theta_{F,E}(A) = \int_F B(\chi_E A) d\mu.$$

If $A \geq 0$, then $\chi_E A \geq 0$, so $B(\chi_E A) \geq 0$ and hence $\Theta_{F,E}(A) \geq 0$. Thus $\Theta_{F,E}$ is positive. Since B preserves the integral,

$$\Theta_{X,E}(A) = \int_X B(\chi_E A) d\mu = \int_X \chi_E A d\mu = \mu(E)A.$$

Also

$$\mu(E)A - \Theta_{F,E}(A) = \Theta_{X \setminus F, E}(A) \geq 0 \quad (A \geq 0).$$

By Lemma 1, there is a scalar $a_E(F) \in [0, \mu(E)]$ such that

$$\Theta_{F,E} = a_E(F) \text{id}_{M_d}.$$

For fixed E , the set function $F \mapsto a_E(F)$ is a finite positive measure: countable additivity follows from countable additivity of the operator integral applied, for instance, to $A = I_d$. Moreover $a_E(F) = 0$ whenever $\mu(F) = 0$. Let

$$h_E = \frac{da_E}{d\mu} \in L^1(X, \mu)_+$$

be the Radon–Nikodym density. Then for all $F \in \Sigma$ and $A \in M_d$,

$$\int_F B(\chi_E A) d\mu = a_E(F)A = \int_F h_E A d\mu.$$

Since M_d is finite-dimensional, equality of all integrals over measurable sets is equality almost everywhere. Thus

$$B(\chi_E A) = h_E A \quad \text{a.e.}$$

Define $S\chi_E = h_E$, and extend linearly to scalar simple functions. The previous paragraph gives

$$B(sA) = S(s)A$$

for every scalar simple function s and every $A \in M_d$. Positivity of B gives positivity of S on simple functions. Taking $A = I_d$ and using $BI_d = I_d$ gives $S1 = 1$. Taking $F = X$ above gives $\int_X S\chi_E d\mu = \mu(E)$, hence $\int_X Ss d\mu = \int_X s d\mu$ for all simple s . Consequently S is contractive on scalar simple functions in the compatible L^1 -norm $\int (|\text{Re } s| + |\text{Im } s|) d\mu$, and therefore extends uniquely to a positive integral-preserving operator on $L^1(X, \mu)$. The identities $S1 = 1$ and $\int Ss d\mu = \int s d\mu$ pass to the extension, so S is classically bistochastic.

Finally let $f \in L^1(X, \mu)$ and $A \in M_d$. Choose scalar simple functions $s_n \rightarrow f$ in $L^1(X, \mu)$. The source paper proves boundedness of multiplication by fixed matrices on $L^1_{\mathcal{H}}(X, \mu I_{\mathcal{H}})$ [1, Proposition 4.4], and B is contractive [1, Proposition 5.4]. Hence

$$B(s_n A) \rightarrow B(fA), \quad S(s_n)A \rightarrow S(f)A$$

in $L^1_{\mathcal{H}}(X, \mu I_d)$, and the identity passes to the limit:

$$B(fA) = S(f)A.$$

For a general $F \in L^1_{\mathbb{C}^d}(X, \mu I_d)$, write $F = \sum_{i,j=1}^d f_{ij} E_{ij}$ with scalar $f_{ij} \in L^1(X, \mu)$; this is the finite-dimensional instance of the coordinate description used in [1, Theorem 5.5]. Then

$$B(F) = \sum_{i,j=1}^d S(f_{ij}) E_{ij}.$$

Thus B is exactly the entrywise extension of S . Uniqueness of S is immediate from $B(fI_d) = S(f)I_d$. $\square$

Limitations

The source sentence is stated for separable Hilbert spaces. The theorem above uses finite dimensionality twice: in the matrix order-interval lemma and in the coordinate/density step extending from simple matrix-valued functions to the whole $L^1_{\mathbb{C}^d}$ space. Thus this packet gives a full no-exotic answer for finite-dimensional quantum systems, but leaves the infinite-dimensional case open.

Verification and novelty check

No computational verification is required. The proof reduces the problem to Lemma 1, which is checked directly on matrix units, and to the standard Radon–Nikodym argument for scalar transition measures.

Local run indexes were searched for arXiv id 2005.00005, the title keywords, and terms including “bistochastic”, “quantum random variables”, “majorization”, and “operator algebra”. No existing packet or attempt covered this result. A bounded web search on 2026-06-21 for exact phrases around “We have no example of a bistochastic operator” and “ $L^1_{\mathcal{H}}(X, \mu I_{\mathcal{H}})$ bistochastic” found no duplicate answer.

Human review should focus on the finite-dimensional order-interval lemma and on the passage from the scalar measures a_E to the classical operator S .

References

- [1] S. Plosker and B. Ramsey, *Bistochastic operators and quantum random variables*, arXiv:2005.00005, 2020.